NOT MEASUREMENT SENSITIVE

MIL-T-62145E
16 January 1990
SUPERSEDING
MIL-T-62145D
9 September 1986

MILITARY SPECIFICATION

TRUCKS: ELEVATING BODY, DIESEL AND GASOLINE ENGINE DRIVEN, 21,000 TO 39,500 POUNDS GVW, 4x2 AND 6x4, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven four-wheel, two-rear-wheel drive, and six-wheel, four-rear-wheel drive vehicles having minimum gross vehicle weights (GVW) of 21,000 to 39,500 pounds, for normal operating conditions. The vehicles are commercial trucks equipped with stake and platform or van bodies, capable of elevating the body and cargo to a specified height for loading and unloading the cargo into and out of aircraft. The vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The vehicle shall be one of the types, classes and models shown in table I, as specified (see 6.2):

TABLE I. Vehicle classification.

Type	Nomenclature	Class	Model
I	Truck, Stake and Platform, with Conventional Cab	A	4X2
		B	4X2
		C	6X4
		D	6X4
		E	6X4
II	Truck, Van, with Tilt Cab	A	4X2
		B	4X2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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1.2.1 Class. The class of the vehicle shall be determined by the minimum gross vehicle weight (GVW) rating as follows:

Class A	21,000
Class B	28,000
Class C	34,500
Class D	39,500
Class E	39,500 (with extended height).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	A-A-393	- Extinguisher, Fire, Dry-Chemical (Hand Portable).
	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	CCC-C-419	- Cloth, Duck, Cotton, Unbleached Plied Yarns, Army and Numbered.

MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
	MIL-B-4792	- Bumper, Rubber, Duplex Round.
*	MIL-C-20696	- Cloth, Coated, Polyester or Nylon, Waterproof.
	MIL-P-27443	- Pallets, Cargo, Aircraft, Type HCU-6/E, HCU-12/E, and HCU-10/C.
*	MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification for.
	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

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STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
MS51317	- Light, Warning, Vehicular, Rotating, D.C.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the European Tyre and Rim Technical Organisation publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles (DoD adopted).
J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings for Six and Eight Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
J706	- Rating of Winches.
J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
J1349	- Engine Power Test Code – Spark Ignition and Diesel.

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(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

(Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.10, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.10, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis components shall be made. All chassis items shall be as represented and rated in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, identification markings, and data plates shall be in accordance with MIL-STD-1223. The exterior color and identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2). When specified (see 6.2), concealed markings shall be furnished.

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* 3.1.1.1.1 Caution and instructions plates. Caution plates shall be furnished and mounted in conspicuous locations in the driver's compartment and at the operator's control stations. Plates showing a schematic of all valves, valve controls, equipment, and complete instructions for operating the body and equipment shall be fastened near the hoist controls in the vehicle cab and near the hoist controls on the body. A body lighting schematic plate shall be furnished. Plates shall conform to composition A (class l or 2) or composition C of MIL-P-514.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in manual transmissions and rear axles shall be of the permanent magnet type.

3.1.1.4 Wood treatment. Wood shall be treated in accordance with MIL-STD-1223.

3.1.1.5 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the chassis manufacturer’s standard single center mounted eye or pin shall be mounted on the front of the vehicle. When specified, in addition, towing devices shall be mounted on the rear of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.6 Wheel splash and stone throw protection. Rear wheels shall have rigid (constructed of metal or rubber with metal trim) splash shields in the front of the rear wheels and rubber mud flaps to the rear of the rear wheels. Splash shields and mud flaps shall be removable and shall be fastened to the chassis. When tilt cabs are furnished, rubber mud flaps shall be provided to the rear of the front wheels. Splash and stone throw protection shall be in accordance with SAE J682. A metal strip, not less than 1/8 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1-1/4 inch wide by 3/32 inch thick shall be furnished at each bolt.

* 3.1.1.7 Trailer towing package. When specified (see 6.2), a trailer towing package shall be furnished. The trailer towing package shall consist of a pintle, safety chain attachment devices, a lighting receptacle, and associated reinforcements and wiring, and shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51118-1. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 4,000 pounds and a horizontal drawbar load of not less than 40,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than four

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inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The attachment devices shall be capable of accommodating a standard grab hook for a 3/8-inch chain (2-13/16 inches wide, 47/64 inch thick, 31/64-inch throat width). The lighting receptacle, conforming to SAE J560, with its conductors connected and color-coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.7.1 Pintle height. When specified (see 6.2), the pintle height, measured to the centerline at the pintle, shall be 28 inches (± 2 inches). The pintle shall be installed with bracket reinforcements to transfer pintle loads directly to the chassis rails. The installation shall be capable of withstanding towed loads equal to the GVW of the truck without permanent deformation.

3.1.1.8 Trailer lighting cable. When specified (see 6.2), a trailer lighting cable, conforming to SAE J1067, shall be furnished. The lighting cable shall be coiled, and shall be not less than 110 inches long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.1.1.9 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with a spark arrester.

* 3.1.1.10 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with Environmental Protection Agency (EPA) Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

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* 3.2.3 Sound level. The cab interior sound level shall be no greater than 84db(A). The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis, cab and body, with all attachments, accessories, and equipment; and a full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. The minimum gross vehicle weight (GVW) shall consist of curb weight; operator weight (computed at 175 pounds); and a payload, distributed as specified in 3.2.6, to provide not less than the minimum GVW specified in table II. In addition, the actual GVW rating of the vehicle furnished shall exceed the minimum GVW specified in table II when required to provide a payload capacity not less than that specified in table II.

TABLE II. Minimum GVW and payload.

Vehicle class	Minimum GVW (pounds)	Minimum payload (pounds)
A	21,000	6,000
B	28,000	10,000
C	34,000	12,000
D	39,500	18,000
E	39,500	20,600

3.2.6 Weight distribution. The distribution of GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the vehicle body load area.

* 3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity.

3.2.8 Overall width. The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

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* 3.2.10 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades specified in table III at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when a multi-speed axle or auxiliary transmission is furnished, with the axle or auxiliary transmission in high speed ratio. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

TABLE III. Gradeability.

Vehicle model	Percent grade
4x2	1.00
6x4	0.50

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide not more than the vehicle speed specified (mph) in table IV for the corresponding number of available forward speeds.

TABLE IV. Vehicle low speed requirement.

Number of forward speeds	Maximum low speed (mph)
5	4.5
7 or 8	3.5
9 or 10	3.0

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

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3.3.3 Subframe, hoist, and body performance.

3.3.3.1 Rated lifting capacity. The rated lifting capacity of the hoist shall be not less than the sum of the weight of the body plus one and one half times the payload specified in table II.

3.3.3.2 Stability. The body shall be stable and rigid throughout the entire elevation and retraction of the hoist mechanism, both unloaded and loaded with the test load specified in table V. In the fully elevated position, with the test load, the body floor shall not deflect more than 1-1/2 inches, measured from the floor to the top of the subframe. The vehicle in any elevated position with the test load specified in table V, shall show no evidence of any tire or outrigger departing from the ground.

TABLE V. Height and test load.

Type	Class	GVW (pounds)	Height (inches)	Test Load (pounds)
I	A	21,000	156	12,000
I	B	28,000	178	20,000
I	C	34,000	222	24,000
I	D	39,500	156	28,000
I	E	39,500	198	30,900
II	A	21,000	156	12,000
II	B	28,000	178	20,000

3.3.3.3 Height. The hoist shall be capable of raising the body with the test load specified in table V from rest position to the minimum height specified in table V. The height shall be measured from the ground to the body floor.

3.3.3.4 Manual locking device. A manual locking device shall be provided to positively retain the body and hoist at a raised height of not less than 96 inches measured from ground level to the body floor.

3.3.3.5 Operating speeds. The hoist and the body shall be capable of elevating and retracting both with the specified payload (see table II) and without payload at a rate of not less than 15 feet per minute (fpm) (10 fpm when retracting without payload), and not more than 20 fpm, with the vehicle engine at idle speed, from the time the load is first imposed on the hydraulic system. The hydraulic fluid shall be stabilized at ambient air temperature prior to the performance of the test.

* 3.3.3.6 Position stability. When the vehicle is on level ground, the hoist hydraulic system shall hold the body loaded with the payload specified in table II, in any elevated position with not more than a 1/8-inch vertical settling in 2 hours. The hydraulic fluid shall be stabilized at ambient air temperature prior to the performance of the test.

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3.3.3.7 Unbalanced load. With one quarter of the payload specified in table II located in any one corner of the body floor, the hoist shall be capable of raising and lowering the body without evidence of binding, deformation, or bending of the hoist mechanism. The body, with the unbalanced load, shall not deflect more than 1-1/2 inches, as specified in 3.3.3.2. No tire or outrigger shall depart from the ground.

3.3.3.8 Wind resistance. The vehicle, both with payload (see table II) and unloaded, at the fully elevated position on a 3 percent sidegrade, without being leveled, shall be capable of withstanding 30 mph winds and wind gusts up to 50 mph from any direction without overturning, without permanent deformation of any component and without deflecting more than 1-1/2 inches as specified in 3.3.3.2. The wind load shall be calculated with the maximum side area perpendicular to the wind force, and shall be measured in accordance with the following:

(a) At the body and hoisting mechanism, when the body is elevated to its maximum specified height, in the most critical position that would induce tipping.
(b) Calculations without outriggers or with the outriggers up shall consider spring and tire deflection of the truck chassis.
(c) The resistant moment shall be greater than the wind load. No tire or outrigger shall depart from the ground.

3.3.3.9 Static load. The body and hoist shall be capable of sustaining a static load equal to three times the payload specified in table II for classes A, B and C and two times the payload specified in table II for classes D and E; when in the any elevated position.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

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* 3.4.1.2 Gasoline. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, a motor rating of 83 and an average octane rating of 87 at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operators manual. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow ready verification of coolant level.

* 3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. For diesel driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, a hydraulic fluid reservoir heater, and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. The heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof, three-pronged (two power plus one ground) male connector at the other end.

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Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with the vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower, coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed.
(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening.
(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than l0°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.
(d) Hydraulic fluid reservoir heater shall have a capacity to maintain the hydraulic fluid at a temperature of not less than l0°F during vehicle exposure in an ambient temperature as low as minus 60°F, and shall embody a thermostat to limit the temperature of the hydraulic fluid to not more than 75°F.
(e) A fuel warmer or pre-heater shall be provided for diesel engines to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt starting system, with 12-volt direct current (dc) lighting system and not less than a 60-ampere alternator shall be furnished. Engine starting equipment shall include an ether starting system, a glow plug or electric grid heater. If an ether starting system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. Measured shot type ether systems shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for an ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

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3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt direct current (dc) ignition system shall be furnished. An alternator of not less than a 40-ampere rated capacity and which provides not less than 16 amperes of dc output at normal engine idle speed shall be furnished.

3.4.2.3 Battery. Each battery shall be of 12-volt potential. The total reserve capacity rating and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than that specified in table VI. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE VI. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel (less than 220 gross horsepower)	320	1,200
Diesel (220 gross horsepower and over)	480	1,740
Gasoline	100	450

3.4.2.4 Lighting. All vehicle lights and reflectors shall be in accordance with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surfaces of the rub rails (unless recessed and fully protected) or on vehicle bumpers. Rear identification lamps, marker lamps and reflectors shall be mounted on the elevating part of the body.

* 3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have visible and audible type flash indicators.

3.4.2.6 Cab spotlights. Two sealed beam spotlights of not less than a 4-1/2 inch diameter shall be furnished. The spotlights shall be mounted on each side of the cab, operable and adjustable from within the cab.

3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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3.4.3.1 Air cleaner. An air cleaner shall be furnished.

3.4.3.2 Fuel tanks. The fuel tank(s) for class A and B vehicles, shall be not less than 50 gallons total capacity (43 gallons for vehicles destined for California); and for class C, D and E vehicles, shall be not less than 90 gallons total capacity. When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel-intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. If vertical exhaust mufflers are furnished, a full exhaust heat shield shall be furnished to protect personnel loading cargo from the front of the body. Vertical exhaust systems shall be provided with a hinged rain cap. A spark arrester shall be furnished, except for vehicles with turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

* 3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be fully shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

* 3.4.5 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with two power takeoff openings conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.1 Speeds. Vehicles with manual transmission shall be provided with the number of vehicle forward speeds specified in table VII. When more than five forward speeds are required, a multispeed axle or an auxiliary transmission shall be furnished. Gear ratios in the transmission and other gear selection equipment shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide the vehicle performance specified in 3.3.1 through 3.3.1.3.

TABLB VII. Minimum number of vehicle forward speeds.

Vehicle model	No. of forward speeds
4X2	5
6X4	9

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* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the class, model vehicle, and engine furnished, with clutch torque capacity exceeding maximum delivered engine torque. The clutch lining shall be asbestos free.

* 3.4.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The transmission shall include a hydraulic torque convertor and not less than four forward gear ratios for 4 x 2 models and five forward gear ratios for 6 x 4 models. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with an SAE J704 power takeoff opening. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

* 3.4.5.4 Power take-off. A power takeoff shall be provided. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution decal or plate reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed, readily visible to the seated driver. The decal or plate shall conform to type I or type II, class 1 of MIL-M-43719 or composition A (class l or 2) or composition C of MIL-P-514.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

* 3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type and class vehicle furnished. The vehicle shall have a heavy duty main frame or frame with reinforcements. The heavy duty frame shall have frame rails of a greater section modulus than the manufacturer's standard for the class vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. Reinforcements shall extend at least from the rear of the front-suspension, rear hanger bracket to the front of the rear-spring, front hanger bracket. Reinforcement shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the load imposed, with body loaded to required GVW and in any elevated position. The chassis frame rails shall not project beyond the rear end of the body.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to required GVW. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels. Suspension system ratings shall equal or exceed the loads imposed during all operating and test conditions specified herein.

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* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to required GVW and during all operating conditions. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Rear bogie (6x4 model). The rear bogie of model 6x4 shall be of the four-wheel type, complete with axles, springs, torque rods, and all other necessary parts. The bogie shall be provided with a means permitting differential action between the two axles, and a manually controlled lockout assuring equal power to each rear axle. Lockout control shall be located in the truck cab. Axle gear ratios shall provide the performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Two-speed axle. When specified (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. Gear ratios shall provide performance specified in 3.3.1 through 3.3.1.3.

3.4.9.3 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle for model 4x2 and on either rear axle on model 6x4. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in procurement documents, multi-piece rims shall not be furnished. Rims and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all wheels on each vehicle. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Tires shall be of the tubeless type with highway tread. Steel belted radial type or when specified (see 6.2), bias ply tires shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the

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vehicle loaded to specified GVW. Tires and tire size designation systems shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

3.4.10.2 Inner tubes. When tube type tires are specified in procurement documents, inner tubes shall be of heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of spare wheel assembly using only the tools specified in 3.4.16.1.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance shall be furnished. Clearance shall be in accordance with SAE J683. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

3.4.11.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7-1/4 cubic feet per minute (cfm) for class A and B vehicles and not less than 12 cfm for class C, D and E vehicles

* (b) Air storage reservoir(s), each tank equipped with drain cock, and with safety and check valves between compressor and last reservoir tank

(c) Foot control, suspended or treadle type

(d) Air control valves

(e) Air pressure gage, visible to the driver

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(f) Low-air-pressure warning, visible and audible
(g) Automatic moisture ejector
(h) Service brake stop lamp switch
* (i) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes.

3.4.11.2 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capacity of removing a minimum of 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.3 Trailer brake control system. When specified (see 6.2), a trailer brake control system shall be furnished in addition to the components specified in 3.4.11.1. Components and accessories shall be mounted on the vehicle chassis. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
(f) Air connectors for trailer with SAE J318 gladhand couplers mounted at the rear of the vehicle, loaded to prevent interference with trailer
(g) Two SAE J844 coiled air hoses, each not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of the hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.4 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

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* 3.4.12 Cab, type I. The chassis manufacturer shall furnish any type of his standard or optional non tilting full-width cab for type I stake and platform vehicles. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. Unless otherwise specified, the cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger's seat. When specified (see 6.2), an individual adjustable driver's seat and an individual passenger seat shall be provided. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be installed. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab.

* 3.4.12.1 Cab, type II. Type II vans shall be furnished with a tilt type cab with a tilting and locking mechanism, and shall be as specified in 3.4.12. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. Provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield. Otherwise, the cab shall conform to 3.4.12.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by either air or electric motor(s).

3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided on each vehicle.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. A rear bumper shall be furnished (see 3.6.13 and 3.7.12).

* 3.4.16 Stowage space. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools shall conform to 3.5.7.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle, and wheel-nut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat.

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The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator, or voltmeter
(c) Fuel gage
(d) Oil pressure gage or red indicator warning light
* (e) Engine temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
(i) Dual sunvisors
(j) Front door mounted armrest on driver's side
(k) Driver's compartment ventilator other than window
(l) Tachometer.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 25 square inches of reflective area. The mirrors shall have not less than two supporting arms.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be provided.

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3.4.22 Fire extinguisher. When specified (see 6.2), two ten-pound, dry chemical portable fire extinguishers shall be furnished. The fire extinguishers shall be of the gas cartridge-operated type for class B and C fires. The fire extinguishers shall be in accordance with type II, class 2, size 10, of A-A-393, having a UL classification of 30B:C. Brackets for two extinguishers shall be furnished, whether fire extinguishers are provided or not.

3.4.23 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the vehicle cab instrument panel or in the engine compartment in a readable location.

3.4.24 Back-up alarm. A back-up alarm shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

* 3.4.25 Chassis mounted rear winch (type I, stake and platform). When specified (see 6.2), type I stake and platform vehicle shall be furnished with a winch assembly, mounted on chassis frame at the rear of the cab, forward of the body. The winch shall be powered by the manufacturer's standard power takeoff or shall be hydraulically driven. The winch shall conform to SAE J706. The winch shall be of the single drum type suitable for hoisting purposes and shall have one forward and one reverse speed. The winch shall have a rated single line pull capacity of not less than 20,000 pounds on the bare drum. The winch shall be wound with not less than 185 feet of 5/8-inch diameter, preformed 6X37, improved plow steel, wire strand core wire rope and shall be equipped with an end chain and hook. An adjustable, automatic safety brake shall be provided. The winch shall be equipped with a 4-way roller guide assembly. Angles of the winch drive line U-joints shall be not more than 16-1/2 degrees. Two sets of winch controls shall be furnished. Each set shall include the following: throttle control; engine master clutch control (manual transmission only); controls for starting, stopping, and reversing the winch; and power takeoff control. One set of controls shall be located in the cab and shall be so located as to provide no interference with the entrance or exit of the driver. The second set of controls shall be mounted on the curbside at the rear. The second set of controls shall be recessed under the body, accessible with the operator standing on the ground at the vehicle curbside. All controls shall be identified.

3.4.25.1 Winch drum guard. A winch drum guard shall be furnished. The winch drum guard shall restrict the cable to the area between the drum flanges. The guard shall consist of not less than 1/4-inch vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 3/8 inch by 1-1/4 inches, shall be welded to the vertical side plates. Three bars shall be

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located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not more than one-half the specified cable diameter.

3.4.26 Body mounted rear winch (type I stake and platform). When specified (see 6.2), type I stake and platform vehicle shall be furnished with a winch assembly mounted on the front of the body floor. The winch shall be hydraulically driven. The winch shall be fully protected against damage from cargo. The winch shall conform to SAE J706. The winch shall be of the single drum type and shall have one forward and one reverse speed and a drum clutch to permit free spooling of the cable. The winch and wire rope shall have a single line pull rating on the bare drum of not less than 10,000 pounds. The winch shall be wound with not less than 150 feet of improved plow steel, preformed regular lay, seale construction wire rope with fiber core and shall be equipped with an end chain and hook. Properly identified operating controls shall be located on or near the front bulkhead. The hydraulic hose connected to the hydraulic drive motor shall be coiled and shall be capable of extending and retracting without snagging and without any appreciable force being applied on the hose fittings as the body is elevated and retracted.

3.5 Body.

3.5.1 Construction. The subframe, hoist and body shall be designed, engineered, and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to the assembly shall be positioned for easy inspection, ready maintenance, and possible replacement and shall have maximum protection against damage afforded by the configuration of the subframe, hoist, and body assembly. The body shall be attached to the subframe. Rotating bearing surfaces which are subject to continuous wear shall be provided with replaceable bearings and bushings. Lubrication fittings shall be provided on all rotating parts. Bearing surfaces shall be adequate to resist torsional moment when inherent in the application.

3.5.1.1 Subframe, hoist and body components. The subframe, hoist and body structural components shall have a calculated design stress such that the yield point of the materials used shall not be exceeded by an induced stress caused by the weight of the components plus the stress due to three times the payload specified in table II imposed on the body floor when the vehicle body is in any of its attainable elevated positions. The materials used shall meet all applicable performance requirements specified herein. Alinement shall be maintained under all operating conditions, without applying stress to any attachment. The hoist subframe longitudinal members shall be continuous, without splices for the length of the subframe.

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3.5.1.2 Subframe. The vehicle shall be equipped with a subframe mounted on the vehicle chassis frame. The subframe shall be the length of the body. The subframe shall consist of the elevating structural members, hydraulic components, and cylinder(s).

3.5.1.3 Hoist. The hoist shall be of the scissors type. Hydraulic cylinder(s) shall be provided to exercise the scissors to elevate the body. The hoist arms shall have sufficient bridging for forward and lateral stability. Means shall be provided to prevent the body from shifting during highway travel, when the body is in the retracted position.

3.5.2 Subframe mounting. The subframe shall be secured with U-bolts, twin studs, or J-bolts, and shall include a wood or steel breaker strip.

3.5.2.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than four U-bolts or twin studs per side, having 9/16-inch minimum body diameter with 5/8-inch minimum thread diameter. Tie-plates shall be at least 1/2 inch thick. The vehicle chassis frame shall be braced, using wood blocks at each mounting point unless the mounting point is located at a full depth subframe crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt, and shall be of a width to assure retention. Two shear bolts shall be provided, one bolted to each side of the rear portion of the subframe to maintain alinement on the vehicle chassis. Forward mounting bolts shall be located to the rear of the tapered portion of the breaker strips.

3.5.2.2 Brackets. When brackets are used, they shall be bolted to the web of the frame rails. The mounting brackets shall provide a means of drawing down the subframe on the chassis rails. Provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to the chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components. Spring-loaded hold-down devices may be used to secure the chassis to the subframe and to provide independent action between the chassis and the subframe.

3.5.2.3 J-bolts. When J-bolts are used, one J-bolt shall be located near each point where a subframe crossmember intersects the vertical plane of the chassis frame rails, to the rear of the tapered section of the breaker strips (see 3.5.2.4). J-bolts shall be not less than 3/8 inch in diameter. J-bolts shall encircle the lower chassis frame flange and shall be securely attached to the subframe. Two shear bolts as specified in 3.5.2.1 shall be provided. Provisions shall be made to prevent lateral shifting of the breaker strips.

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3.5.2.4 Breaker strips. Wood breaker strips shall be of hardwood or dense southern yellow pine, of not less than 1-1/16 inches in finished thickness. The breaker strips shall be installed between the longitudinal sills and the vehicle chassis frame and shall have a taper of 1 inch in 18 inches at the forward end. When a chassis mounted rear winch is specified, the breaker strip shall terminate rearward of the winch and the taper is not required.

3.5.3 Fenders. Rear fenders with heavy rubber splash guards shall be provided over the rear wheels. The fenders shall be of not less than 14 gage (0.0747 inch) sheet steel. A fender flange around the wheel cutout shall be furnished. The fenders shall be at least the width of the rear dual tires. The fenders shall be supported by the chassis frame and shall be demountable with hand tools.

3.5.4 Electrical wiring. All body wiring shall be in conduit. Splices are not acceptable. All wiring ends shall have staked lugs or connectors. Wiring shall be in accordance with SAE J1292. Wiring shall be coded. Floodlights and interior body lights shall be furnished with three-way switches at the lamps and on the vehicle chassis cab instrument panel.

3.5.5 Body lighting.

* 3.5.5.1 Floodlights. Two sealed beam, 12-volt, 100-watt, universal mounted floodlights shall be furnished for night cargo-transfer operations. The floodlights shall be a minimum of 5-1/2 inches in diameter and shall be corrosion-resistant. For type I stake and platform vehicles, the floodlights shall be located at the rear of the platform and shall be mounted in a position to prevent breakage. For type II van type vehicles, the floodlights shall be located on the outside front wall of the body on either side of the front roll-up door, shall be operable from the front cab platform (see 3.7.11) and shall not protrude above the body roof line. When specified (see 6.2), two additional floodlights capable of being rotated 360 degrees horizontally shall be mounted on the front of the type I stake and platform body or on the rear of type II van body.

* 3.5.5.2 Interior lighting, type II van vehicles. Eight interior lights shall be furnished on type II vans. Four lamps shall be on both sides of the body. The lights shall be recessed within the side walls. The lights shall be rated at a minimum of 25 watts per lamp. The lamps shall have guards to prevent damage. The guards shall be hinged at the top and shall be secured to prevent movement.

* 3.5.5.3 Flashing warning lights. Two red flashing warning lights shall be furnished. The warning lights shall flash whenever the body is being elevated or retracted. One light shall be located on the front of the bulkhead and one at the rear for type I stake and platform and one light shall be located over the front door and one over the rear door for type II van. Both lights shall be outside the van.

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3.5.5.4 Rotating warning light. An amber, rotating, vehicular warning light in accordance with MS51317-4 shall be provided. The rotating light shall be mounted on top of the cab for type I stake and platform and over the front door on the roof of the elevating body for type II van. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch.

* 3.5.5.5 Cable. A color coded, multiconductor cable, enclosed in metal or plastic flexible loom, shall be routed up along the scissors mechanism to provide for electrical power and controls for the elevated body. The routing shall provide an ample radius at the center of the scissors mechanism to minimize damage from flexing. The cable end connections and fastenings to the scissors mechanism shall provide for ready replacement of the cable.

3.5.6 Hydraulic system.

3.5.6.1 Hydraulic pump. A hydraulic pump with compatible pressure and flow capacities to operate all hydraulic components and accessories shall be furnished. The hydraulic pump shall be driven by a heavy duty power takeoff unit attached to the truck-chassis transmission. A split-shaft power takeoff is not acceptable. Chain drives are not acceptable. The hoist pump shall be of the gear type. The pump continuous pressure rating shall be not less than that recommended by the hoist and cylinder manufacturer(s), with an intermittent rating not less than 500 pounds per square inch (psi) over the continuous rating. The system hydraulic pressure shall be limited to not more than 2500 psi under any operating condition. The pump system shall include a built-in adjustable pressure relief valve(s) or unloading valve(s), with a range of adjustment which allow the operating and the test loads specified herein. The hydraulic pump shall be capable of operating in ambient temperatures from minus 40°F to plus 125°F.

3.5.6.2 Neutral switch. A neutral switch shall be furnished. The neutral switch shall be activated when the transmission is in neutral to operate the pump. When an automatic transmission is provided, a brake lock shall be provided that will not permit the pump to operate unless the brake lock is activated.

3.5.6.3 Automatic throttle actuator. An automatic hydraulic or electric throttle actuator shall be furnished. The actuator shall automatically increase the truck engine speed when the pump is being loaded, and automatically decrease the speed of the truck engine when no load is being imposed.

3.5.6.4 Hydraulic reservoir. A hydraulic reservoir shall be furnished. The reservoir shall be equipped with a drain, a filler opening cap attached to the reservoir by a corrosion-resistant chain, and internal baffles to prevent excessive surging of the fluid. A dip stick or sight gage

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shall be provided to enable the level of fluid to be observed when the lift body is in its lowered position. A tank vent shall be installed and shall prevent weather and foreign matter from entering the reservoir. Provisions shall be made for inspecting and cleaning the reservoir without loss of fluid in the system. The reservoir shall be protected to prevent damage. The reservoir shall be provided with a photoetched, corrosion-resistant plate to indicate the type of hydraulic fluid to be used. The plate shall be installed near the filler cap. When specified (see 3.4.1.8(d)), the reservoir shall be equipped with a heater.

3.5.6.4.1 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

3.5.6.4.2 Oil filter and strainer. A full flow type 10-micron discharge oil filter with a capacity not less than three times the maximum pump operating flow rate shall be provided in the return line to the reservoir. A 100 mesh suction line strainer with a capacity of not less than two times the pump flow rate shall be provided in or near the hydraulic reservoir. The return line oil filter shall be equipped with a condition indicator gage and a 15 psi internal bypass. The gage shall be so located as to be easily readable with the body in the down position. Both the filter and the suction strainer shall be accessible for cleaning or for replacement of the filter cartridge.

3.5.6.5 Hydraulic fluid temperatures. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal damage and faulty operation of hydraulic valves and controls. Excessive heating of the fluid (more than 55°F above ambient in an ambient of 125°F) shall not be acceptable under any operating condition.

3.5.6.6 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, visual inspection for failures, and maximum protection from damage during operation of the body assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings, adaptions to flared type fittings, or tapered pipe thread with dry seal fits shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to prevent vibration and chafing. High pressure hoses which are located in areas where chafing may occur shall be covered with spiral type hose guards. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and for completely draining all hydraulic lines, when necessary. A manual type bleeder shall also be provided as on-vehicle equipment on closed-center systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable. A pressure gage

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shall be provided on the discharge side of the hydraulic pump. Accumulators shall not be furnished. The system shall be designed to a safety factor of 3 for normal operating conditions.

3.5.6.6.1 Hydraulic valving. An in-line relief valve shall be provided, fully adjustable, and compatible with the working loads and test loads specified herein. The valve shall have provisions for a safety lock wire on the adjustment portion of the valve to permit sealing and locking at the proper working pressure after test. The valve shall be sealed and locked at the proper setting after testing. A normally closed, 12-volt dc, solenoid operated check valve or valves shall be provided to lock the cylinder(s) in the event of a failure in any other portion of the hydraulic system. If the check valve is not integral with the cylinder, a minimum number of fittings shall be used between the check valve and the cylinder, with a burst strength of at least three times the maximum operating pressure. A normally open 12-volt dc, solenoid operated, two-position, two-way valve with pilot assist or a three-position four-way valve with both cylinder ports blocked shall be provided to raise and lower the body. The high pressure hydraulic system shall be provided with a gage port to accommodate a 2-1/2 inch face, 0-3000 psi, glycerin filled, stem-mounted gage. The gage, when installed, shall be easily visible with the body in the down position. The gage need not be furnished.

3.5.6.7 Hydraulic cylinders. The lift mechanism shall be actuated by single or dual cylinders. When dual cylinders are used, they shall be connected to a hydraulic manifold in a manner to assure equal pressure and simultaneous operation of both cylinders. Cylinders may be single, multiple telescoping, or other special design. All seals, rings, gaskets, and other related components shall be designed for easy removal and replacement. Metal nonflexing hydraulic lines shall be used where flexing is not required during operation. An armored or otherwise protected flexible hose shall be used where flexing is required. Provisions shall be made for bleeding the system at all necessary points without removing or loosening any component of the system. The cylinder(s) shall be positioned to prevent the cylinder(s) from extending above the body floor. The hoist cylinder(s) shall be furnished with automatic cushioning of their stroke when approaching the fully extended and fully retracted positions of the body. The hydraulic system shall be designed to prevent hydraulic transients. Automatic pilot check valves shall be provided to prevent body descent in the event of a hydraulic, electrical, or power source failure. A manual device or valve shall be incorporated into the system to enable lowering of the body in case of an electrical or power source failure. The device will bleed the fluid back into the reservoir. The device shall provide metering characteristics.

3.5.6.7.1 Hydraulic cylinder valving and manifold. A check valve or a ball type shut-off valve shall be provided between the reservoir and the return line filter to enable the shut off of the oil supply to facilitate changing the element in the filter. A fixed orifice device or an

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adjustable throttle valve (flow control) shall be located as close as practicable to the lift cylinder(s) to limit and control the downspeed. The throttle valve shall be rated for not less than three times the working pressure and shall be equipped with a lock screw or jam nut to hold the desired preset. When two lift cylinders are provided, a manifold shall distribute equal pressure and flow rate and shall provide for simultaneous operation of the cylinders.

3.5.6.8 Controls. Hoist controls shall be electrically operated and shall be of the dead man type. Means shall be provided for manual operation of the controls in the event of an electrical failure. The controls shall be metered to provide gradual starting and stopping of the hoist and to eliminate shock due to start and stop.

3.5.6.9 Control location. Two sets of controls for raising, stopping and lowering the body shall be provided. One set shall be readily accessible to an operator standing on the ground and shall be at the rear end of the body for type I stake and platform, and at the front end of the body for type II van. The second set shall be in the cab, readily accessible to an operator in the driver's seat. The control being operated shall override the idle control.

3.5.6.10 Outriggers. Class E vehicles shall be provided with four outriggers. For all other classes, if required to insure adequate stability with relation to the ground under the conditions specified in 3.3.3.2, 3.3.3.7, and 3.3.3.8, outriggers shall be provided. Outriggers shall be double acting and hydraulically operated. Interlocks shall be provided, allowing the body to raise to the maximum stable height and the vehicle to move (without outriggers) before the vehicle must be stopped and the outriggers must be extended, to enable the body to raise to its full height. Additional interlocks shall be provided to prevent the transmission from engaging a forward or reverse gear when the outriggers are down. Outrigger controls shall be installed in a readily accessible location allowing operation from the driver’s seat. Outrigger cylinders, hydraulic fittings, hoses and pilot check valves shall be protected by metal guards and shall be accessible for service. An automatic pilot check valve shall be provided for each outrigger to block the hydraulic fluid in the cylinder and to prevent movement of the outriggers in the event of a damaged hydraulic line. Each outrigger shall be provided with an automatic, positive, spring-actuated lock, or an automatic pilot check valve which will hold the outrigger in the retracted position without settling and which may be manually released. Each outrigger shall be provided with a ground pad. Ground pads shall be of the hinged or fixed type, each not less than 64 square inches in area.

* 3.5.7 Tool storage cabinet. A tool storage cabinet shall be furnished. The storage cabinet shall be capable of stowing the tools as specified in 3.4.16.1 (if tools are required or not), load restraint devices, anti-skid chains, power cables, and emergency reflective triangles. The storage cabinet shall be weatherproof. The storage cabinet shall be fabricated of steel not less than

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10 gage (0.1345 inch) thick. A bottom hinged door shall be furnished, with corrosion-resistant stay chains to provide a shelf when horizontal. A hasp with latch and provisions for a padlock shall be furnished. The inside corners shall be reinforced. The cabinet shall be supported and bolted to the chassis frame rails by use of corrosion-resistant bolts and nuts. Gussets shall be provided. The cabinet shall be capable of being removed by the use of wrenches. The cabinet shall be electric arc welded.

3.5.8 Telescopic ladder. When specified (see 6.2), a telescopic ladder for access to the body loadspace on type I and type II vehicles shall be furnished. Type I stake and platform ladder shall be located at the streetside rear of the body. Type II van ladder shall be located at the curbside front for access onto the front overcab platform. The ladder shall be constructed of aluminum members and shall be designed to prevent injury when ascending or descending and when the body is being elevated or retracted. The ladder shall not extend over the top of the roof of the body in any elevated or retracted position of the body. Each ladder rung shall be capable of accepting a 600-pound load imposed at its center without evidence of deformation.

* 3.5.9 Safety. All equipment or exposed portions of the equipment which create extreme temperatures or that are subject to damage from inclement weather, and all rotating or reciprocating or elevating parts, which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded. All sharp or protruding edges shall be eliminated, enclosed or properly guarded.

3.5.9.1 Safety devices. The following safety devices shall be included:

(a) A device to prevent sudden lowering of the body through failure of any component of the mechanical, hydraulic, or electrical system other than the hydraulic cylinder(s)
(b) A manually operated locking device to positively retain the body at not less than 8 feet above ground level to the body floor for maintenance purposes
(c) An electrical cut-off switch and positive stops to retain the body at any height at which the body is located
(d) A device with actuating controls and with an indicator installed in the vehicle cab, to preclude movement of the vehicle when the body operating system is being activated, and to prevent operation of the body operating system when the vehicle is in motion. The device shall be controlled from the vehicle cab and shall be accessible to the operator when in a seated position. The device shall be installed so that the control cannot be operated unless the clutch pedal is depressed or when an automatic transmission is furnished, when the detent is placed in the drive position.

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3.5.9.2 Warning signs. A warning sign specifying the payload capacity of the body shall be provided on the body visible to a driver entering the driver's cab door. Warning signs shall be furnished in critical locations reading: "KEEP HANDS AND FEET CLEAR OF LIFT WHILE LIFT IS IN OPERATION."

3.5.9.3 Highway warning kit. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and stowed in the tool storage cabinet.

3.5.10 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.5.11 Lubrication fittings. Lubrication fittings accessibility located shall be furnished where moving parts normally require lubrication.

3.6 Type I, stake and platform body. Type I stake and platform body shall have the minimum dimensions specified in table VIII.

TABLE VIII. Type I dimensions (inches).

Designation	Vehicle class A	B	C	D	E
Cab-to-axle without chassis mounted winch	100	120	144	100	138
Cab-to-axle dimension with chassis mounted winch	114	138	161	118	--
Overall platform length	168	192	240	168	230
Overall width (plus 1 inch, minus 1 inch)	95	95	95	95	95
Interior width (plus 1 inch, minus 0)	84	84	84	84	90
Rack height (plus 1 inch, minus 0)	42	42	42	42	42

3.6.1 Construction. In addition to the requirements of 3.5 through 3.5.11, the body shall consist of a steel tread plate floor, side racks, tailgate, solid steel front, roof bows, and a tarpaulin. When a chassis mounted rear winch assembly is specified, the body shall be mounted rearward of the winch assembly. When the side racks and the front assembly are removed and the tailgate lowered, and the platform elevated above the vehicle cab, the platform shall present an entirely flat, unobstructed, horizontal surface.

3.6.2 Body frame. Body framing shall be a completely welded structure with members of minimum gage thickness specified in table IX for carbon steel; high tensile steel 50,000 psi minimum yield strength may be furnished in two gages lighter weight in accordance with US Standard gage sizes. There shall be not less than 10 full-width crossmembers on class A vehicles; not less than 12 full-width crossmembers on class B and C vehicles; and not less than 13 full-width crossmembers on class D and E vehicles, including ends and stub crossmembers as required for proper spacing over the axle(s). Rub rail shall be attached using continuous welds.

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Not less than 4 additional full width crossmembers on class C vehicles and not less than 5 additional full width crossmembers on class D and E vehicles shall be provided in the area of the rear bogie. The additional crossmembers may be joined by welding to the normally located crossmembers in the rear bogie area. Crossmembers shall be reinforced by gusset plates or brackets at points of attachment to longitudinal sills. Contact edges of crossmembers with longitudinal sills, and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge length. Longitudinal sills shall be constructed of structural steel channels or formed channels. Formed channel sills shall be reinforced within the sill, at each crossmember or body mounting point, with formed channel reinforcements. Frame members shall be of a continuous length without splices or joints.

TABLE IX. Framing gages.

Designation	US Standard gage No.	Equivalent inches
Crossmembers	10	0.1345
Side and end rails	10	0.1345
Longitudinal sills	8	0.1644
Reinforcements	8	0.1644
Rack posts	11	0.1196

* 3.6.3 Flooring. Flooring shall be safety steel tread plate not less than 1/4 inch in thickness exclusive of the projections. A medium tread plate pattern shall be furnished. When the body floor is constructed in two or more pieces, continuous seam welds having full penetration shall be provided the full length of each seam.

* 3.6.4 Front bulkhead. A full-width removable front bulkhead shall be furnished. The bulkhead shall be not less than 42 inches in height, measured from the platform floor. The bulkhead shall be formed or reinforced for rigidity. The bulkhead shall be provided with a screen window in alinement with the rear window of the truck cab. The screen shall be corrosion-resistant, of not more than No. 6 mesh, number 20 wire, having a square opening of 0.132 inch. The screen shall be enclosed in a metal frame and secured by screws to the bulkhead. The screen window shall be covered on the inside with a No. 2 mesh, number 16 wire, having a square opening of 0.437 inch. The bulkhead stakes shall be on the outside of the payload area and shall be bolted to the bulkhead. The bulkhead shall be secured in stake pockets and to the side sections with draw-down type fasteners. Openings for the chassis mounted winch line or body mounted winch hydraulic lines, if a winch is furnished, shall be provided. The bulkhead shall be capable of withstanding a horizontal static load equal to one half the payload capacity of the vehicle without permanent distortion of the bulkhead or its mountings.

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3.6.5 Stake pockets. Stake pockets shall be furnished and shall be of the protruding type, providing drainage. The pockets shall be located inside the rub rail. The pockets shall be welded to the rub rail and the body.

3.6.6 Side racks. There shall be not less than three removable rack sections on the curbside and not less than three removable rack sections on the streetside. Racks shall be equipped with steel fasteners. Body hardware attaching bolts shall be not less than 2 inches from the ends of rack slats. Upright posts shall be steel sections. Rack slats shall be of oak, hickory, maple, or ash of not less than 25/32 inch finished thickness. Width of individual slats shall be manufacturer's standard, providing that the total of slat widths is not less than 60 percent of the total rack height. There shall be not less than four slats per rack. Slat edges shall be beveled. End corners shall have the top and bottom of the slat ends rounded with not less than 1/2-inch minimum radii. Slats shall be riveted or bolted to the inside (load side) of the upright posts, with rivet or bolt heads against the slat. Each side rack section at the rear of the body shall be provided with at least one draw-down type fastening, equipped with a locking nut, to secure the racks in place. The upright posts shall fit the stake pockets with a clearance not to exceed 3/16 inch.

3.6.7 Roof bows. Unless otherwise specified (see 6.2), roof bows shall be provided. Roof bows shall be spaced on not less than 20-inch and not more than 30 inch centers. Roof bow sockets shall be provided on the outside of the rack assemblies. The sockets shall accept a roof bow having a cross sectional dimension of 1 inch by 2-1/2 inches. Sockets shall be provided with a bow stop and drainage shall be provided. Tie-down devices for securing the bows shall be furnished. The height of the loadspace, measured at the longitudinal centerline from the floor to the roof bow, shall be not less than 88 inches. The edges of the roof bows shall be blended and smooth to preclude abrasion of the tarpaulin.

* 3.6.8 Tarpaulin. Unless otherwise specified (see 6.2), a tarpaulin shall be furnished. The tarpaulin and curtains shall be similar in color to the exterior color of the vehicle. The tarpaulin shall enclose the body and shall reach the floor level with the bows in place. The tarpaulin shall have reinforced corners, eyelets, and reinforcements over each bow to withstand chafing. The tarpaulin shall be fabricated of number 8 cotton duck conforming to type I of CCC-C-419, or of vinyl coated nylon conforming to type II, class 2 of MIL-C-20696. The material shall be water repellant and fire-resistant. The rear curtain shall be of the roll-up type. The front curtain shall have a window which shall be not less than 12 inches by 24 inches in size and shall aline with the front bulkhead window and the rear window of the vehicle cab. Tarpaulin tie-down devices shall be provided on the same centers as the roof bows.

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* 3.6.9 Tailgate. Unless a hydraulic tailgate is specified (see 3.6.10), a tailgate conforming to the following shall be provided. The tailgate shall be of aluminum. The tailgate shall be transversely centered and its width shall be the same as the body loadspace. The height of the tailgate in the travel position shall be not less than 40 inches measured from the centerline of the tailgate hinge pivot point. The tailgate shall be hinged at the bottom and supported with duck-covered, corrosion-resistant stay chains on each side of the tailgate. The top or the surface facing the loadspace shall have a non-skid tread surface. The tailgate shall be of the ramping type, tapering down to 3/8 inch plus 1/4 inch minus 0 thickness at the aircraft floor level, starting approximately 9 inches from the edge. The bottom side of the ramp shall have a smooth bearing surface to prevent damage to aircraft floors. The tailgate shall be capable of withstanding a 5,000 pound load, with the tailgate in a horizontal position and supported by its hinges and the stay chains. The tailgate shall be capable of being placed and secured in the vertical downward, horizontal, and vertical up positions. Rubber guards shall be furnished on the tailgate at those points which may come into contact with a loading dock when the tailgate is completely up, extended horizontally, or hanging downward.

* 3.6.10 Hydraulic tailgate. When specified (see 6.2), a hydraulically powered, hinged tailgate shall be furnished. The height of the tailgate shall be not less than 42 inches measured from the body floor and the width shall be not less than the width of the body loadspace. The tailgate shall be hinged at the bottom. The tailgate shall be provided with duck-covered corrosion-resistant safety chains on each side of the tailgate, for support in the horizontal position. The hydraulic system shall be a constant pressure hydraulic system to provide power to raise the tailgate to the vertical up position and to lower the tailgate to the horizontal position or vertical down position in smooth operations. Means shall be provided to secure the tailgate in the vertical plane up, in the horizontal plane and in the vertical plane down. The tailgate surface which faces the loadspace shall have a non-skid tread surface. The bottom side of the tailgate shall have a smooth bearing surface to prevent damage to aircraft floors. Rubber guards shall be furnished on the tailgate at those points which may come in contact with a loading dock with the tailgate either vertical or horizontal. The tailgate shall be of the ramping type, tapering down to 3/8 inch plus 1/4 minus 0 thickness at the aircraft floor level, starting approximately 9 inches from the edge, to facilitate loading and unloading with wheeled loaders and handlers. When the tailgate is in the loading (horizontal) position, the clearance between the rear edge of the vehicle and the tailgate shall be not more than 3/4 inch. The tailgate shall be capable of withstanding a load of 5,000 pounds with the tailgate in the horizontal position and supported by its hinges, the hydraulic system and the safety chains. The tailgate shall be capable of being held in either the vertical or horizontal position when the vehicle is traveling. The hydraulic cylinders shall be provided with flow restrictors in the downport of the cylinders to prevent the tailgate from falling in the event of hydraulic system failure. Controls shall be mounted near the tailgate to facilitate operation from the ground and from the body with the body in the elevated position. An

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instruction plate shall be positioned near the controls, describing the operation of the hydraulic tailgate. Hydraulic hoses connected to the hydraulic cylinders shall be coiled and shall be capable of extending and retracting with the platform without snagging and without exerting excessive loads on the hose fittings.

* 3.6.11 Rollers. When specified for class E vehicles (see 6.2), removable rollers shall be installed on the body floor to facilitate handling, without damage, two 10,000 pound capacity, 300 pound HCU-6/E pallets conforming to type I of MIL-P-27443, loaded to capacity. The rollers shall be of a width sufficient to allow 88-inch wide pallets to be loaded onto the truck bed. The rollers shall be capable of being installed and removed by four men, within one hour, using only common hand tools. A means shall be provided on the vehicle to lock the pallets securely to the rollers to prevent movement. The devices that hold the rollers onto the truck bed floor and the pallet lockdown devices shall not protrude above the floor when the rollers have been removed.

3.6.12 Portable ramps. When specified (see 6.2), two portable rear loading ramps shall be furnished. The ramps shall be of aluminum. The ramps shall be not less than 166 inches in length, 9 inches in width, with a tread pattern on the top surface. The ramps shall have a channel section at the upper section. Means shall be provided to anchor the ramp ends to the edge of the platform and to permit varying the spacing in at least 10-inch increments. Each ramp shall be capable of supporting a 4,500-pound imposed load on a span 12 feet in length when supported only at the ends. Storage provisions for the ramps, with cam type tie-downs, shall be furnished.

3.6.13 Rear bumper. The rear of the chassis frame shall be boxed in with material at least equal to the chassis frame material and shall be adequate for use as a bumper. The bumper shall be installed so as to permit the rear tailgate to hang down perpendicular to the truck bed.

3.6.14 Rear step. A step shall be provided on the curbside of the rear of the body to assist personnel entering the cargo space when the body is in its lowered position and the tailgate is in its horizontal position. The step shall be of the stirrup type and shall be installed to prevent damage to the step when the truck is backed up against a loading dock.

3.6.15 Transition plate. When specified (see 6.2), a transition plate shall be furnished to bridge the gap between the tailgate and an aircraft cargo door when the truck is positioned at a 45-degree angle to the aircraft. The plate shall be trapezoidal in shape, approximately 7 feet wide, 4 feet long on the short side and 11 feet long on the long side (see figure 1). The edge shall be tapered to facilitate the movement of wheeled loaders and handlers. The transition plate shall be attachable to the tailgate in a manner to provide a relatively smooth surface

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(see detail "A", figure 1). The plate shall be capable of withstanding on imposed load of not less than 4,000 pounds. The transition plate shall be manufactured in sections with locking edges to prevent spreading during loading, or otherwise manufactured to facilitate the handling of the plate.

3.7 Type II, van body. Type II van body shall have the minimum dimensions specified in table X. In addition to the requirements of 3.5 through 3.5.11, the body shall consist of an overhead cab platform with a roll-up front door to a van body interior and a roll-up rear door.

TABLE X. Type II dimensions (inches).

Designation	Vehicle class	
	A	B
Cab-to-trunnion	100	120
Length (inside)	168	192
Height (inside) (with roll-up doors)	78	78
Width (inside)	89	89

3.7.1 Body subframing. Body subframing shall be of all-steel welded construction, braced at all points of stress. There shall be not less than 12 full width crossmembers, including ends, of full channel or equivalent section, pressed steel construction, attached to longitudinal sills by welds, and at front, rear, and every second intervening crossmember by gussets. When subframing is by welded assembly, angle gussets, crossmembers, longitudinal sills, and welded reinforcement (when used) shall be joined by weld for not less than 50 percent of the length of the contact edge(s). Spacing of crossmember shall be equal, ahead of, and to the rear of the wheels. Longitudinal sills shall be constructed of structural steel channel or structural I-beam or reinforced formed channels. Formed channel sills shall be reinforced within the sill, at each crossmember or body mounting point with formed channel reinforcements. Crossmembers, gussets, formed longitudinal sills, and reinforcements shall be not less than 10 gage (0.1345 inch) steel. Side and front rails (out-rails) shall be not less than 12 gage (0.1046 inch) steel. The design of the body subframe shall provide tire chain clearance and shall permit vehicle operation at speeds up to 58 mph over improved roads, when loaded to the required GVW, without evidence of permanent deformation.

3.7.2 Body framing. Side and front post construction shall be extruded aluminum and shall be not less than 1.26 inches in depth and 0. 430 pound per foot. Side and front wall posts shall be spaced on not more than 12-inch maximum centers. Front corner post construction shall be extruded aluminum and shall be not less than 1.0 inch in depth and 0.482 pound per foot. Rear corner posts shall be not less than 12 gage (0.1046 inch) steel.

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3.7.3 Body interior. The interior of the body walls, including the ceiling, shall be lined with not less than 1/4-inch exterior grade plywood. The interior radius corners may be aluminum or fiberglass lined. The ceiling shall be insulated with not less than l-inch fiberglass.

3.7.4 Flooring. The flooring shall be safety aluminum tread plate not less than 1/4 inch thick, exclusive of the tread height. The floor shall be of one-piece construction without welds or seams or, when one piece construction is not obtainable, two-piece construction may be provided when the two pieces are of equal size. When two-piece construction is furnished, an automatic continuous seam weld having full penetration shall be furnished or when the seam is hand welded, the underside of the seam shall be reinforced with a 3-inch wide by not less than 0.1875-inch thick aluminum butt plate welded between the crossmembers. The floor shall be provided with an aluminum coving or an aluminum rolled radius around the perimeter of the wall, except at the door openings. The coving or rolled radius shall be the same thickness as the floor, shall have a vertical height of at least 4 inches, but need not have a tread pattern unless the coving and the floor are of the same material. The coving shall be continuously welded when two pieces are furnished. The coving shall be secured with corrosion-resistant fasteners to the wall. The floor shall be provided with drains.

3.7.5 Interior rope ties. Interior rope ties, either fold down lashing rings, rope cleats, or rod type, shall be secured to wall frame members. A minimum of 24 rope ties shall be arranged in three evenly spaced tiers, four rope ties per tier on each side wall; the first tier located approximately 5-1/2 inches above the floor line; the second tier at 1/3 of wall height; and the third tier located at 2/3 of wall height. The rope ties shall not project into the cargo area beyond the wall lining.

3.7.6 Body exterior. The roof assembly shall be constructed to insure drainage. Roof support bow construction shall be aluminum and shall be not less than 1.0625 inches in depth and 0.394 pounds per foot. Roof bows shall be spaced on not more than 12-inch centers. The roof panel shall be of one-piece aluminum not less than 0.040 inch in thickness. The roof panel shall be stretched and secured. The roof panel shall not flutter or “tin can.” Body panels shall be not less than 0.051 inch in thickness. Body side panels shall be removable to facilitate body repair. All roof and body seams and joints shall be weatherproof. All body and roof rails shall be extruded aluminum and shall have a maximum radius of 4.5 inches. Rub rails shall be provided along each side of the body. Rub rails shall be permanently attached to or shall become a part of the exterior of the body at the floor line. Drip-moldings shall be furnished over each window and over the front and real doors.

3.7.7 Windows. Two windows shall be furnished on each side of the body. Window casings, screens, components, parts, and accessories shall be constructed of corrosion-resistant aluminum. Corrosion-resistant properties shall be inherent in the chemical composition of the aluminum. The windows shall be waterproof. The windows shall provide a clear glass opening

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of not less than 12 inches in width by 18 inches in height. The glass shall be safety plate grade. The windows shall be located midway between the roof and the longitudinal centerline of the body.

3.7.8 Window guard. Window guards shall be furnished on the inside of each window. The window guards shall be enclosed in an aluminum frame, and hinged to the body. A cam type latch shall be furnished. Screening for the windows shall be aluminum and shall have openings of not less than 0.437 inch.

3.7.9 Roll-up overhead rear door. A roll-up overhead rear door shall be provided. The door shall provide a clear opening of not less than 80 inches in width and not less than 72 inches in height. The door shall be of the sectional type having not less than six sections. Door sections shall incorporate joints of a tapered tongue-and-groove or shiplap type. The door shall be plastic-covered or aluminum-covered 3/4-inch marine plywood construction. Door section corrosion-resistant hinges shall be mounted on the inside of the door, not less than four hinges at each section joint. The door track shall be manufacturer's standard corrosion-resistant type equipped with a positive stop at the end of the track. Door rollers, a counter-balance unit, and cables of corrosion-resistant construction shall be provided. The door shall be weather-tight and shall close not less than 1/4 inch below the floor level. A heavy duty corrosion-resistant cam-operated lever type key lock shall be provided. One heavy duty corrosion-resistant grab handle, closed type, shall be provided on the left outside bottom of the door. One nylon double loop pull-down strap not less than 2 inches wide and 12 inches in length shall be provided and located adjacent to the door lock. The door shall be capable of being locked and unlocked from inside and outside the body.

3.7.10 Roll-up overhead front door. A roll-up overhead door, centrally located at the front of the body, shall be furnished. The door shall provide a clear opening of not less than 48 inches in width and 72 inches in height. The door shall meet the requirements of 3.7.9, except as specified herein. A structural frame member covered with flattened expanded metal shall be installed in back of the vehicle chassis cab in alinement with the front door of the body. The expanded metal shall be not less than 16 gage (0.0598 inch) and shall have an opening of not more than 1-1/2 inches, measured across the narrow part of the opening. The perimeter of the expanded metal shall be provided with a moulding to cover the raw edges. Frame height shall be up to the chassis cab rear window.

* 3.7.11 Front platform. A platform shall be furnished for front-end loading. The platform shall extend far enough forward so that the vehicle operator, seated in the driver's seat in the vehicle cab, can see the front edge of the platform when the platform is at the extreme elevated height specified in table V. The platform shall be the same width as the elevating body. The

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platform shall be capable of supporting a 1,000-pound load distributed over any one square foot of area. With the elevating body retracted and resting on its subframe, the platform shall be supported by a structural installation located at the back of the vehicle cab, and the platform shall clear the vehicle cab roof by not less than 6 inches. When the body is being elevated, the body shall pick up the platform at the body floor level and raise it with the body, maintaining both floor levels in the same plane. The threshold of the elevating body or the platform shall be provided with a means to prevent the feet of operating personnel in the elevating body from being engaged in the platform over the vehicle cab when the elevating body is ascending. Pads shall be provided for a hydraulic jack. Pads shall be capable of accepting the loads imposed. The jack shall raise the platform to enable servicing of the vehicle.

* 3.7.11.1 Platform bumper. A platform bumper shall be furnished on the front and the two sides of the platform. The bumpers shall be in accordance with MIL-B-4792, except the bumper shall be a single, hollow round section not less than 7-inches in outside diameter and not more than 5-1/8 inches in inside diameter. Means shall be provided to arrest the elevation of the body when the bumpers contact any external object and to lower the body until the contact is cleared. The force of the contact required shall not exceed 50 pounds.

3.7.11.2 Platform floor. The platform floor shall be one- or two-piece, aluminum safety tread plate not less than 1/4 inch thick, exclusive of tread height. Two-piece floor construction shall conform to 3.7.4.

* 3.7.11.3 Platform guard rails. Removable guard rails shall be furnished. The guard rails shall enclose the curbside, streetside, and front of the platform. (The front guard rail will be removed for loading and unloading.) The guard rails shall be constructed of aluminum structural members and shall be not less than 42 inches in height, measured from the floor to the top of the guard rails. The side guard rails shall be secured to the body and to the floor with quick-clamp type fasteners, and shall be capable of being removed without the use of tools. The front guard rail shall be secured to the side guard rails with quick-clamp type fasteners and shall be capable of being removed without the use of tools. Hangers shall be provided in the body for stowage of the guard rails. Stowage provisions shall include means of clamping the guard rails down when not in use.

3.7.12 Rear step bumper. A spring loaded rear bumper shall be furnished. A safety tread step, integrated with the rear bumper, shall be provided at the rear of the body to assist personnel in entering the body interior when the body is in the lowered position. The step shall extend across the full width of the body and shall be not less than 10 inches deep.

3.8 Welding. Adequate preparation of metal prior to welding of the body subframe, hoist, elevating body, and components shall be provided. The surface of parts to be welded shall be free of rust, scale, paint, grease, and other foreign matter. Welding rods shall be compatible with

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the metal being welded. All critical arc welds shall be magnetic particle or dye penetrant inspected prior to painting. The following defects shall not be acceptable:

(a) Welds which are ground flush with the metal
(b) Slug or plug welding
(c) Undersize of fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments and vertical down welding
(d) Unremoved slag and spatter
(e) Cracks
(f) Porosity.

Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.8.1 and 3.8.2.

3.8.1 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.8.2 Factors. The factors listed below, which are applicable, shall be included in the recorded welding procedure:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type and thickness
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range, and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat
(j) Welding process
(k) Shielding gas
(l) Method of cleaning.

3.8.3 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the body, hoist, subframe, and components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

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3.9 Stress analysis. A stress analysis for the body shall be performed. The stress analysis shall include the weight of the chassis and of all major body items permanently installed on or within the vehicle, and shall show the location of the center of gravity of such components relative to the vehicle axles (raised and lowered positions). In addition, the analysis shall show the value of stresses in the vehicle chassis frame and the subframe, and at the trunnions of the cylinders in the lowest position and when raised to the highest position. The analysis shall verify that the yield point of the material used is not exceeded under the conditions, the safety factors, burst pressures and working pressures specified herein (see 3.5.1.1, 3.5.6.6 and 3.3.3 through 3.3.3.9, etc).

3.10 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alinement of lights; adjustment of the engine, electrical and brake systems; filling and charging of battery; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment, with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume, and servicing of windshield washer reservoir with water and appropriate additives.

3.11 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the

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contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspection.

* 4.3.1 Welding procedure and workmanship specimen evaluation. Prior to the start of fabrication of the subframe and hoist and components and assemblies thereof, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures (see 3.8.1 and 3.8.2).

4.3.1.1 Macro-etching. Workmanship specimens shall be cross-sectioned and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

* 4.3.2 In-process inspection of body and components. The first complete assembled body and components (subframe and hoist, and their components and assemblies) shall receive an in-process inspection by the Government prior to prime painting. The inspection shall be conducted to evaluate workmanship and to determine conformance to structural requirements. The vehicle shall be tested in accordance with 4.3.2.1 through 4.3.2.3 prior to prime painting, and dye penetrant or magnetic particle testing shall be performed on all critical areas to inspect for crack indications under dynamic and static conditions. Any evidence of cracks or permanent deformation shall be cause for rejection of the vehicle.

4.3.2.1 Dynamic conditions. With the test load specified in table V, the body shall be raised and lowered for at least 1/2 hour while being inspected for cracks and permanent deformation.

4.3.2.2 Static conditions. With the body in its highest position, three times the payload specified in table II for classes A, B, and C and two times the payload for classes D and E shall be placed on the body.

4.3.2.3 Critical areas. At least the following critical areas shall be inspected by the dye penetrant or magnetic particle testing:

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(a) Subframe, trunnion, yoke, body cylinder guides, and attaching areas
(b) Struts, hoisting mechanism
(c) Structural steel of body and subframe
(d) Rear tailgate support areas, type I stake and platform
(e) Front platform support areas, type II van.

4.4 First vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight and GVW. The first production vehicle shall be weighed to determine the curb weight and the distribution of the curb weight on the front and rear axles. The imposed loading on the front and rear axle(s) shall be computed by the contractor and verified by the Government and compared to the capacities of the suspension, axles, and tires to assure these components are of adequate capacity to meet contractual requirements. Distribution of weight shall be computed by using the curb weight, the operator weight at 175 pounds, and a payload to provide the required GVW (see 3.2.5).

4.4.2 Road test prior to operational tests. Prior to the tests in 4.4.4 through 4.4.4.12, the first production vehicle shall be road tested. The vehicle, loaded to required GVW (see 3.2.5) and with all equipment specified herein, shall be driven at speeds up to at least 55 mph, for a minimum distance of 30 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 10 miles of travel at least one sudden stop shall be accomplished. Upon completion of the road test, all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all body and cab doors shall properly open and shut; there shall be no evidence of mud or dust inside type II van and there shall be no evidence of racking of the body.

* 4.4.3 Body treatment and painting. A certification regarding the body cleaning, treating, prime painting and salt spray resistance testing as required by MIL-STD-1223 shall be made to Government representatives at the first production vehicle inspection.

4.4.4 Operational testing.

4.4.4.1 Upsetting and range of movement. With the vehicle resting on approximately level ground and not fastened to any artificial base, the body shall be loaded with the test load specified in table V. The body shall be extended to its full height, lowered to travel position and raised to increments of one third, two thirds, and its full height, and lowered to travel position. The body shall be observed to insure that it stays level and for malfunctions or permanent deformation. The tires and outriggers shall be observed to insure they remain in contact with the ground.

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4.4.4.2 Wind load test. Wind resistance shall be demonstrated with the payload specified in table II and without payload. With the payload, with the body elevated to its maximum height and with the vehicle positioned whereby the greatest area is parallel to the centerline of the vehicle, the wind load, generated by an artificial means or by an equivalent simulated wind, shall be applied to the vehicle from the sides (see 3.3.3.8). The tires and outriggers shall be observed to insure they remain in contact with the ground.

4.4.4.3 Elevating body speed test. With the vehicle resting on approximately level ground and the payload specified in table II uniformly distributed over the body loadspace, the body shall be elevated to its specified height (see table V) and held for 2 minutes, plus or minus 1/2 minute, and lowered to rest position forty cycles, with a 30-minute interval between each 10 cycles. Each cycle equals one complete elevation and retraction of the body. The test shall verify the operating speed requirements of 3.3.3.5 and the temperature requirements of 3.5.6.5. The test shall be performed in an ambient temperature of 75°F plus or minus 10°F. The test shall be repeated for 5 cycles without load to determine conformance to 3.3.3.5 without load.

* 4.4.4.4 Creep test. With the vehicle resting on approximately level ground, and an evenly distributed full payload (see table II) placed on the body, the body shall be extended to its full height and held for not less than two hours. Vertical settling shall be not more than 1/8 inch in two hours (see 3.3.3.6). Hydraulic oil shall be stabilized at ambient temperature before time measurement starts.

4.4.4.5 Unbalanced load test. With the vehicle resting on level ground, one quarter of the payload specified in table II shall be located in any one corner of the body floor. The hoist shall then raise and lower the body ten times without evidence of binding, deformation or bending of the hoist mechanism. The body with the unbalanced load shall not deflect more than 1-1/2 inches measured from the floor to the top of the subframe.

* 4.4.4.6 Tailgate, type I stake and platform. The tailgate on type I stake and platform vehicles shall be lowered and a 5,000 pound load imposed. The tailgate, hinges, chains, chain supports and, when applicable, hydraulic system, shall be capable of accepting this imposed load, as specified.

4.4.4.7 Platform bumper, type II van. The platform bumper on type II van shall stop the body elevation when the bumper contacts an external object and shall lower the body until contact is broken (see 3.7.11.1).

4.4.4.8 Telescopic ladder. The telescopic ladder, when provided, shall be capable of extending and retracting when the body is being exercised (see 3.5.8).

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4.4.4.9 Hoist stability. During all tests, the hoist shall provide sufficient structural strength to prevent shifting laterally or fore and aft, without evidence of undue lateral flexing of the structure and without evidence of wear due to binding of the hoist members.

4.4.4.10 Controls. Controls shall be operated and shall meet or exceed the requirements of 3.5.6.8, 3.5.6.9 and, when applicable, 3.4.25, 3.4.26, 3.5.6.10 and 3.6.10.

4.4.4.11 Hydraulic system verification. The hydraulic system shall be thoroughly inspected to assure that the requirements of 3.5.6 through 3.5.6.10 and 3.6.10 are met.

4.4.4.12 Body electrical system. The body electrical system shall be inspected to assure conformance to 3.5.4 through 3.5.5.6.

4.4.5 Servicing and adjustment of the hydraulic system. Servicing and adjustment of the hydraulic system shall be demonstrated as specified in 3.5.6.

4.4.6 Final road test. A final road test shall be performed after the tests of 4.4.4.5 through 4.4.4.12 have been completed. The road test shall be as specified in 4.4.2, except the minimum distance shall be 10 miles.

4.4.6.1 Final operation test. A final operational test shall be performed subsequent to the road test in 4.4.6. The body shall be fully elevated and retracted with a uniformly distributed full payload (see table II) for not less than 30 cycles, with a 30-minute interval between each 10 cycles. The body assembly shall be subjected to, but not limited to, the following checks:

(a) Controls activate when the neutral switch is not activated, when the transmission is in neutral
(b) Controls return to neutral position when the operator releases the control
(c) Both controls conform to 3.5.6.8 and 3.5.6.9.
(d) Operate equipment using each set of controls separately
(e) Check to insure that the entire system can be operated smoothly and without jerking motions in any position
(f) Check to determine whether the body can be safely lowered, from maximum reach position, with rated payload, with the hydraulic pump and engine not operating
(g) Check the hydraulic system, including the pump at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.

4.5 Production sample. Upon acceptance of the first production vehicle, the vehicle shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped

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on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrective action has been taken.

4.7 Inspection of production vehicles. The contractor's inspection system shall as a minimum assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required in 3.9 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. Vehicles shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are for nontactical use by the Government in loading and unloading aircraft.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and model required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for painting and marking (see 3.1.1.1).
(e) Concealed markings, if required (see 3.1.1.1).
(f) Rustproofing, if required (see 3.1.1.2).
(g) Tropical rustproofing, if required (see 3.1.1.2).
(h) Rear towing devices, in addition, if required (see 3.1.1.5).
(i) Trailer towing package, if required (see 3.1.1.7).
(j) Pintle height, if required (see 3.1.1.7.1).
(k) A trailer lighting cable, if required (see 3.1.1.8).
* (l) Satisfactory operation on JP-4, JP-5 and JP-8 fuel, if required (see 3.4.1.1).

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(m) Gasoline engine, if required (see 3.4.1.2).
(n) Power plant heaters, if required (see 3.4.1.8).
(o) If a battery heater is not required (see 3.4.1.8).
(p) Automatic or semiautomatic transmission, if required (see 3.4.5.3).
(q) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(r) Two-speed axle, if required (see 3.4.9.2).
(s) Traction control, if required (see 3.4.9.3).
(t) Wide base tires, if required (see 3.4.10).
(u) Disc type wheels, if required (see 3.4.10).
(v) Bias ply tires, if required (see 3.4.10.1).
(w) A carrier, if required (see 3.4.10.3).
(x) A spare wheel or rim, if required (see 3.4.10.4).
(y) Spare tire assembly, if required (see 3.4.10.5).
(z) A trailer brake control system, if required (see 3.4.11.3).
(aa) Brake controls for use from a towing vehicle, if required (see 3.4.11.4).
(ab) Tilting hood and fender assembly, if required (see 3.4.12).
(ac) Individual driver's and passenger's seats, if required (see 3.4.12).
(ad) Tools, if required (see 3.4.16.1).
(ae) Air operated horn in addition to an electric horn, if required (see 3.4.21).
(af) Fire extinguishers, if required (see 3.4.22).
(ag) Chassis mounted rear winch for type I, if required (see 3.4.25).
(ah) Body mounted rear winch for type I, if required (see 3.4.26).
(ai) Additional floodlights, if required (see 3.5.5.1).
(aj) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.5.6.4.1).
(ak) Telescopic ladder, if required (see 3.5.8).
(al) If roof bows are not required (see 3.6.7).
(am) If a tarpaulin is not required (see 3.6.8).
(an) Hydraulic tailgate, if required (see 3.6.10).
(ao) Rollers, if required (see 3.6.11).
(ap) Portable ramps, if required (see 3.6.12).
(aq) Transition plate, if required (see 3.6.15).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contractor. The SAE Work Sheet item 1 should include vehicle model number, engine model number, and vehicle type, class and model. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

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Table 2	-	Altitude Factor. 1.00.
Table 3	-	Rolling Factor. 1.613.
Table 4	-	Area Factor. 0.173.
Table 5	-	Velocity Factor. 250.0.
Table 6	-	Altitude Factor. 1.00.
Table 7	-	Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	-	Grade Factor. 0.75.
Table 8A	-	Correction Factor. Not required.
Table 9	-	Road Factor. 0.0.

6.4 Subject term (key word) listing.

Aircraft loading
Commercial vehicle
First production vehicle
Gross vehicle weight
Hydraulics
Nontactical vehicle
Outriggers, hydraulic
Pallet rollers
Stability
Stress analysis
Truck, nontactical
Truck, stake and platform
Truck, van
Wind resistance.

6.5 Changes from previous issue. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for the inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

Custodians:
Army – AT
Navy – YD
Air Force – 99

Preparing Activity:
Army – AT

Project No. 2320-0430

Reviewing Activities:
Navy – MC
Air Force – 84

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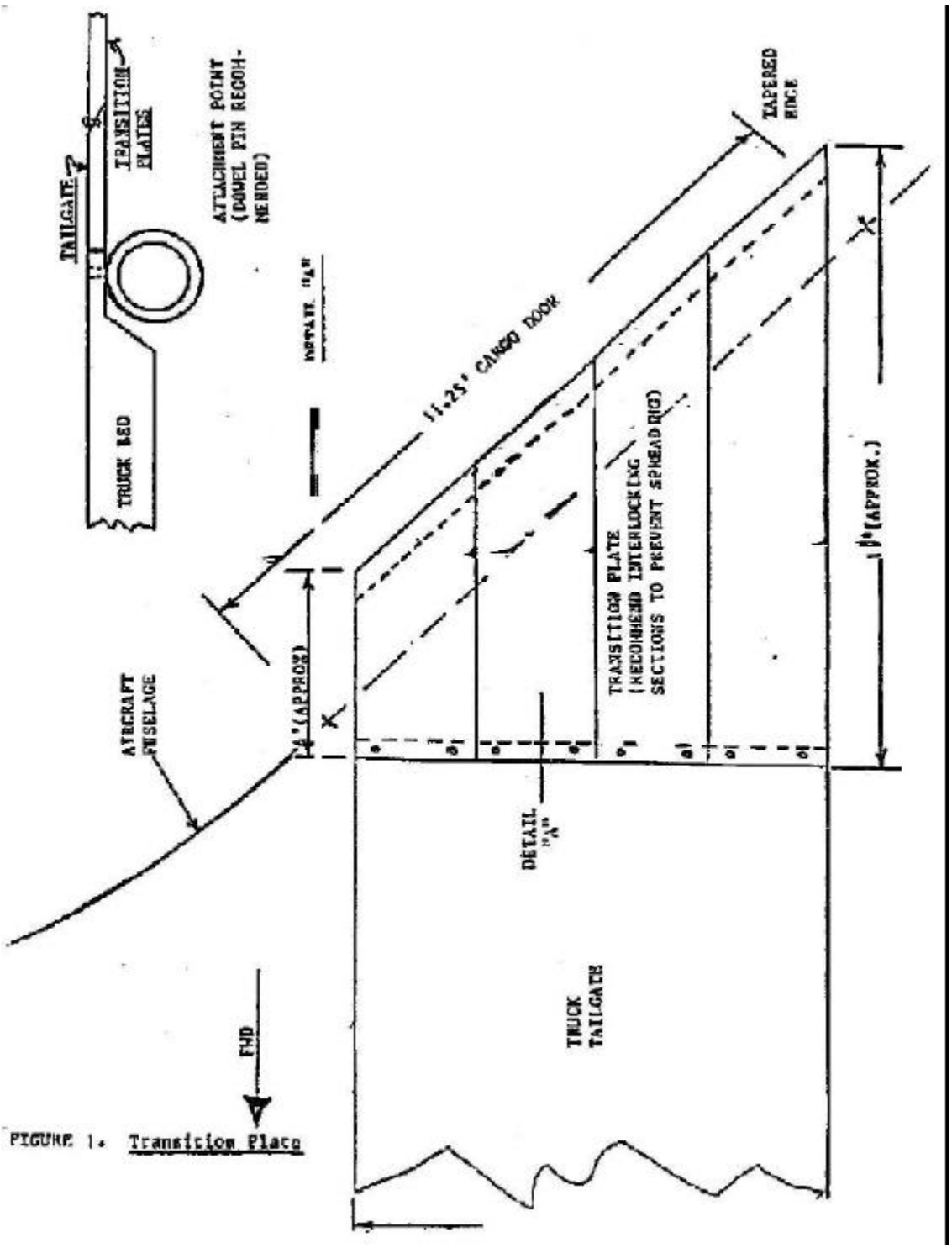


FIGURE 1. Transition Plate

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